Practice Questions

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Score: 100% Passing Score: 80%

[redacted]

Question 1: ✔ Correct

Which pins in an RJ45 connector are used to transmit data when used on a 100BaseT Ethernet network? (Select two.)

- [x] Pin 1
- [x] Pin 2
- [ ] Pin 3
- [ ] Pin 4
- [ ] Pin 5
- [ ] Pin 6
- [ ] Pin 7
- [ ] Pin 8

EXPLANATION

On a 100BaseT network cable, the RJ45 pin-outs are as follows:

- Pin 1: Tx+
- Pin 2: Tx-
- Pin 3: Rx+
- Pin 4: Unused
- Pin 5: Unused
- Pin 6: Rx-
- Pin 7: Unused
- Pin 8: Unused

For a 100BaseT cable, Pins 1 and 2 are used to transmit data; pins 3 and 6 are used to receive data.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 2: ✔ Correct

Which recommendation should you follow while using 110 blocks for connecting Cat 5 and higher data cables?

- ➡ Keep wire pairs twisted up to within one-half of an inch of the connector.
- Connect wires using the T568B standard.
- Use C-5 connectors.
- Connect wires using the T568A standard.

EXPLANATION

While using Cat5 (or higher) wiring, preserve the twists in each wire pair to within one-half of an inch of the connecting block.

Use C-4 connectors to connect four pairs of wires. When connecting data wires on a 110 block, you typically connect wires in the following order:

- White wire with a blue stripe followed by the solid blue wire.
- White wire with an orange stripe followed by the solid orange wire.
- White wire with a green stripe followed by the solid green wire.
- White wire with a brown stripe followed by the solid brown wire.

T568A and T568B are used to connect wires within an RJ45 connector when making drop cables.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 3: ✔ Correct

You are building network cables and attaching RJ45 connectors to each end. Which tool do you need for this task?

- Punch down tool
- Wire puller
- ➡ Crimping tool
- Needle nose pliers

EXPLANATION

You should use a crimping tool designed for RJ45 connectors to attach connectors to UTP cable.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 4: ✔ Correct

You are working with 25 pair wires and 66 blocks. You have pushed the wires onto the 66 block, but now you need to cut off the excess end of each wire.

Which tool should you use?

- Needle nose pliers
- Cable stripper
- Snips
- ➡ Punch down tool

EXPLANATION

Use a punch down tool to push wires into 66 or 110 blocks and cut wires at the same time. The punch down tool has a blade on one side that cuts off the excess wires.

Use snips to cut cables or wires. Use a cable stripper to remove the plastic covering for a cable. When making drop cables or using punch down blocks, do not remove the plastic covering for individual wires.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 5: ✔ Correct

You want to use the T568B standard for adding connectors to your Cat5 cable. Starting with pin 1, which order should you use for the wires within the connector?

- White/blue, blue, white/orange, orange, white/green, green, white/brown, brown
- White/green, green, white/orange, blue, white/blue, orange, white/brown, brown
- ➡ White/orange, orange, white/green, blue, white/blue, green, white/brown, brown
- White/orange, orange, white/green, green, white/blue, blue, white/brown, brown

EXPLANATION

The T568B standard uses the following order of wires in the connector: white/orange, orange, white/green, blue, white/blue, green, white/brown, brown.

The T568A standard switches the green and orange wires (along with their corresponding white wires). Use the order blue-orange-green-brown (BLOG), with the white wire first for connecting wires on a 110 punch down block.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 6: ✔ Correct

Which of the following methods would you use to create a crossover cable?

- Use the T568B standard.
- Use the BLOG convention.
- ➡ Use the T568A standard on one connector and the T568B standard on the other connector.
- Use the T568B standard on one connector and the BLOG convention on the other connector.
- Use the T568A standard on one connector and the BLOG convention on the other connector.
- Use the T568A standard.

EXPLANATION

The easiest way to create a crossover cable is to arrange the wires in the first connector using the T568A standard and arrange the wires in the second connector using the T568B standard. A crossover cable connects the transmit pins on one connector to the receive pins on the other connector (pin 1 to pin 3 and pin 2 to pin 6).

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 7: ✔ Correct

You need to terminate a Cat 6 UTP cable with an RJ45 connector. Your organization's IT policy states that all cable connectors must be wired according to TIA568B standards.

Drag the UTP wire on the left to the correct pin in the RJ45 connector on the right.

Pin 1	Pin 2	Pin 3
✔ White with orange stripe	✔ Solid orange	✔ White with green stripe
Pin 4	Pin 5	Pin 6
✔ Solid blue	✔ White with blue stripe	✔ Solid green
Pin 7	Pin 8	
✔ White with brown stripe	✔ Solid brown	

EXPLANATION

The T568B wiring standard specifies the following pin-out for RJ45 connectors:

- Pin 1: White with orange stripe
- Pin 2: Solid orange
- Pin 3: White with green stripe
- Pin 4: Solid blue
- Pin 5: White with blue stripe
- Pin 6: Solid green
- Pin 7: White with brown stripe
- Pin 8: Solid brown

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 8: ✔ Correct

You need to terminate a Cat 6 UTP cable with an RJ45 connector. Your organization's IT policy states that all cable connectors must be wired according to TIA568A standards.

Drag the UTP wire on the left to the correct pin in the RJ45 connector on the right.

Pin 1 ✔ White with green stripe	Pin 2 ✔ Solid green	Pin 3 ✔ White with orange stripe
Pin 4 ✔ Solid blue	Pin 5 ✔ White with blue stripe	Pin 6 ✔ Solid orange
Pin 7 ✔ White with brown stripe	Pin 8 ✔ Solid brown	

EXPLANATION

The T568A wiring standard specifies the following pin-out for RJ45 connectors:

- Pin 1: White with green stripe
- Pin 2: Solid green
- Pin 3: White with orange stripe
- Pin 4: Solid blue
- Pin 5: White with blue stripe
- Pin 6: Solid orange
- Pin 7: White with brown stripe
- Pin 8: Solid brown

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 9: ✔ Correct

Which of the following is used to terminate individual wires from a 25 pair or 100 pair cable using female RJ45 ports?

- Horizontal cross connect
- ➡ Patch panel
- 110 block
- 66 block

EXPLANATION

A patch panel is a device that typically connects individual stranded wires into female RJ45 connectors. For example, you might connect 4 pairs of wires from a punch down block to a port on the patch panel. On the patch panel, you then connect drop cables (cables with RJ45 connectors) to the patch panel on one end and a computer on the other end.

Use 66 and 110 blocks to connect individual wires within a wiring closet. These punch down blocks connect the individual wires together, but do not terminate in RJ45 connectors.

A horizontal cross connect connects IDFs on the same floor.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 10: ✔ Correct

Which of the following describes the point where the service provider's responsibility to install and maintain wiring and equipment ends, and the customer's responsibility begins?

- Vertical cross connect
- Punch down block
- ➡ Demarc
- IDF

EXPLANATION

When you contract with a local exchange carrier (LEC) for data or telephone services, they install a physical cable and a termination jack onto your premises. The demarcation point (demarc) is the line that marks the boundary between the telco equipment and the private network or telephone system. Typically, the LEC is responsible for all equipment on one side of the demarc, and the customer is responsible for all equipment on the other side of the demarc.

A punch down block is a block used to connect individual copper wires together. While the demarc might terminate in a punch down block, punch down blocks are used within other locations at the customer site.

An intermediate distribution frame (IDF) is a smaller wiring distribution point within a building. IDFs are typically located on each floor directly above the main distribution frame (MDF), although additional IDFs can be added on each floor as necessary. A vertical cross connect connects the IDF to the MDF on a different floor.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 11: ✔ Correct

You have a network that occupies all three floors of a building. The WAN service provider has installed the line for the WAN service into the building in a wiring closet on the main floor. You have a wiring closet on the two remaining floors directly above the wiring closet on the main floor.

What would you use to connect the wiring closets together?

- ➡ Vertical cross connect
- Demarc extension
- Smart jack
- Horizontal cross connect

EXPLANATION

A vertical cross connect joins the main distribution frame (MDF) on the main floor to intermediate distribution frames (IDFs) on upper floors. Cabling runs vertically (up and down) between the MDF and the IDFs.

A horizontal cross connect joins IDFs on the same floor. Cabling runs horizontally (sideways) between the IDFs. A smart jack is a special loopback plug installed at the demarcation point for a WAN service. Technicians at the central office can send diagnostic commands to the smart plug to test connectivity between the central office and the demarc.

A demarc extension extends the demarcation point from its original location to another location within the building. The demarc extension typically consists of a single wire bundle that attaches to the existing demarc and supplies a termination point to a different location. You might need a demarc extension if your network occupies an upper floor of a building. The LEC will typically install the demarc into the MDF on the bottom floor, and you will need to install an extension to place the demarc into the IDF on your floor.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 12: ✔ Correct

Which of the following terms identifies the wiring closet in the basement or a ground floor that typically includes the demarcation point?

- Horizontal cross connect
- ➡ MDF
- Smart jack
- IDF
- 110 block

EXPLANATION

The main distribution frame (MDF) is the main wiring point for a building. The MDF is typically located on the bottom floor or basement. The LEC typically installs the demarc to the MDF. An intermediate distribution frame (IDF) is a smaller wiring distribution point within a building. IDFs are typically located on each floor directly above the MDF, but you can place additional IDFs on each floor as necessary.

A horizontal cross connect joins wiring closets on the same floor. A smart jack is a special loopback plug installed at the demarcation point for a WAN service. Technicians at the central office can send diagnostic commands to the smart plug to test connectivity between the central office and the demarc.

Use 66 and 110 blocks to connect individual wires within a wiring closet.

REFERENCES

LabSim for Network Pro, Section 2.4.

Question 13: ✔ Correct

You have a network that occupies both floors of a building. The WAN service provider has installed the line for the WAN service in a wiring closet on the main floor.

You have a second wiring closet on the second floor directly above the wiring closet that holds the demarc. Which of the following terms describes the closet on the second floor?

- Horizontal cross connect
- MDF
- ➡ IDF
- Vertical cross connect
- Demarc extension

EXPLANATION

An intermediate distribution frame (IDF) is a wiring distribution point within a building that is smaller than the MDF. IDFs are typically located on each floor directly above the MDF, but you can place additional IDFs on each floor as necessary.

The main distribution frame (MDF) is the main wiring point for a building. The MDF is typically located on the bottom floor or basement. The LEC typically installs the demarc to the MDF. A vertical cross connect connects the MDF on the main floor to IDFs on upper floors. Cabling runs vertically (up and down) between the MDF and the IDFs. A horizontal cross connect connects IDFs on the same floor. Cabling runs horizontally (sideways) between the IDFs.

A demarc extension extends the demarcation point from its original location to another location within the building.

REFERENCES

LabSim for Network Pro, Section 2.4.

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